Name:

BUID:

Workshop 4: Subspaces, Dimension, Ranks-Nullity

1. Let A denote the matrix $\begin{bmatrix} 1 & -2 & 19 & 0 & -4 \\ 1 & 0 & 9 & 1 & 1 \\ 1 & -1 & 14 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & 0 & 2 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$

   (a) Determine a basis for Col A. (b) Determine a basis for Nul A. (c) Determine the dimension of Col A.
   (d) Determine the dimension of Nul A.

2. Let A be the matrix $\begin{bmatrix} 1 & -4 & 3 & -3 \\ -2 & 8 & -6 & 7 \end{bmatrix}$.

   (a) Determine a basis for Col A. (b) Determine a basis for Nul A. (c) Determine the dimension of Col A.
   (d) Determine the dimension of Nul A.

3. Determine the coordinate vector $\mathbf{v} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ relative to the basis $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ of $\mathbb{R}^2$ (i.e. determine $[\mathbf{v}]_{\mathcal{B}}$).

4. Determine the coordinate vector $\mathbf{v} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$ relative to the basis $\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$ of a subspace of $\mathbb{R}^2$ (i.e., determine $[\mathbf{v}]_{\mathcal{B}}$).

5. Suppose that $[\mathbf{u}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$ where $\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$. Determine $\mathbf{u}$.

6. Let $A \in \mathbb{R}^{8\times 11}$ be a matrix such that $\dim \operatorname{Nul} A = 5$. Determine rank A.

7. Let $A \in \mathbb{R}^{9\times 15}$ such that A^T has linearly independent columns. Determine $\dim \operatorname{Nul} A$.

8. *(Optional)* Show that if U and V are subspaces of $\mathbb{R}^n$, then so is $U \cap V = \{\mathbf{v} : \mathbf{v} \in U \text{ and } \mathbf{v} \in V\}$.

Write your solutions below and on the back on this page.

Continue your work here.